Curriculum Vitae/Resume
Rupadarshi Ray

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Research interests

My primary research interests include the following areas.

- Hyperbolic manifolds and discrete subgroups of Isom$(\mathbb{R}\boldsymbol{H}^n) \cong O^+(1,n)$, Poincare series, Patterson-Sullivan densities on limit sets of discrete subgroups
- Rigidity of discrete groups/geometric structures: Mostow's rigidity theorem, Sullivan's and Tukia's generalizations
- Deformations of discrete groups/geometric structures: Teichmuller spaces, quasiconformal deformations, Schottky spaces, space of discrete faithful representations of finitely generated groups into Isom$(\mathbb{R}\boldsymbol{H}^n) \cong O^+(1,n)$

My broad interests also include geometric topology and 2 or 3-manifolds, abstract harmonic analysis, symplectic geometry and Hamiltonian flows, representation theory of Lie groups and Lie algebras, etc.

Academic referees

- Arghya Mondal, arghyamondal@iisermohali.ac.in, Assistant Professor, IISER Mohali (MS thesis supervisor)
- Pranab Sardar, psardar@iisermohali.ac.in, Associate Professor, IISER Mohali
- Kapil Hari Paranjape, kapil@iisermohali.ac.in, Professor Emeritus, IISER Mohali
- Vaibhav Vaish, vaibhav@iisermohali.ac.in, Assistant Professor, IISER Mohali
- Soma Maity, somamaity@iisermohali.ac.in, Assistant Professor, IISER Mohali

Higher education and dissertations

Indian Institute of Science Education and Research Mohali (IISER Mohali) 2021-2026
5 years BSMS, graduating with a Masters in Mathematics Mohali, Punjab, India

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Quasiconformal and measurable rigidity of discrete groups . . . . Fall 2025-Spring 2026
Masters thesis under Dr. Arghya Mondal IISER Mohali

TALK Gave a talk surverying the developments in the two generaliations of Mostow's rigidity theorem for my defence

POSTER Presented a poster on Teichmuller spaces of hyperbolic surfaces and Mostow's rigidity theorem

Talks, seminars (co)organized, projects and thesis

Kleinian groups in higher dimensions . . . . Summer 2026-
Research project under Dr. Pranab Sardar

TALK Classical and non-classical Schottky groups
TALK Space of Schottky groups and its boundary

TALK On hyperbolic 3-manifolds and Kleinian groups titled "Our history with hyperbolic 3-manifolds" in the Graduate Students Seminar, IISER Mohali, Summer 2026

Boundary of symmetric spaces . . . . Fall 2025-Spring 2026
Seminar organized by Dr. Arghya Mondal IISER Mohali

TALK On reductive subgroups of $GL(n, \mathbb{R})$ and totally geodesic submanifolds of $P(n, \mathbb{R})$
TALK On the visual and Tits boundaries of CAT(0) symmetric spaces
TALK Counting orbit points of discrete subgroups, Poincare series and critical exponents of discrete subgroups
TALK On the construction of Patterson-Sullivan measures on limit sets
TALK On the proof of Mostow's rigidity theorem by Sullivan using Patterson-Sullivan measures

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TALK On proper actions and properly discontinuous actions on topological spaces, Spring 2026

TALK On deformations and rigidity of geometric structures: Teichmuller spaces and Mostow's rigidity following an article by Lizhen Ji, online, Winter 2025

C^*-algebras . . . . Fall 2025
Seminar (co)organized with graduate students in IISER Mohali IISER Mohali

⋆ Student talks on spectrum of Banach and C^*-algebras following Deitmar-Echterhoff
TALK Proof of Gelfand-Naimark theorem for C^*-algebras

Ergodic theorems for the actions of locally compact amenable groups . . . . Summer 2025
Reading project under Dr. Jotsaroop Kaur IISER Mohali

TALK On singular integral transforms on $\mathbb{R}^n$, the transference principle applied to an inequality with the Hilbert transform
TALK On ergodic transformations, spectral invariants of ergodic transformations
TALK On amenable groups, amenability of compact groups and Abelian groups
TALK On ergodic theorems for measure preserving transforms
TALK On strong continuity of left and right translation on locally compact groups
TALK On ergodic theorems for measure preserving group actions by locally compact Abelian groups

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Meromorphic functions on Riemann surfaces . . . . Summer 2025
Seminar (co)organized IISER Mohali

- ⋆ Introductory talks by Dr. Kapil Hari Paranjape on etale space of sheaf of holomorphic and meromorphic functions, ringed space definition of Riemann surfaces
- ⋆ Student talks on constructing meromorphic functions on Riemann surfaces, Riemann-Roch theorem

TALK On Abel's theorem and Jacobi's inversion theorem

TALK On symplectic manifolds and Hamiltonian flows titled "When does a vector field kill area?" in the Graduate Students Seminar, IISER Mohali, Spring 2025

Courses and reading projects

Selected courses credited

2022 *Geometry of curves and surfaces* following *Pressley* and *do Carmo*

2024 *Knots and braids*: Reidemeister moves, colouring invariants, Jones polynomial, Alexander module and its elementary ideals, braid groups

2024 *Riemannian geometry* following *Jack Lee*

2025 *Arithmetic of elliptic curves*

2025 *Fourier analysis*

2026 *Fuchsian groups*

The following are talks I gave in courses that I have credited.

TALK On computing the genus of modular curves, for the course on *Fuchsian groups*, IISER Mohali, Spring 2026

TALK An elementary proof of Selberg's lemma for $SL(2, \mathbb{C})$, for the course on *Fuchsian groups*, IISER Mohali, Spring 2026

TALK On the construction of Haar measure on locally compact Hausdorff groups: for the course on Fourier analysis, IISER Mohali, Semester 8

TALK On the periods of elliptic curves: for the course on Arithmetic of elliptic curves, IISER Mohali, Semester 8

TALK On Knower's and liar's paradoxes and their formalizations titled "What does a knower and a liar have in common?", course on Philosophy of language, IISER Mohali, Semester 7

Reading projects/self-reading

2023 Audited a course on *Algebraic topology* following Hatcher

2023 Smooth manifolds and differential forms following *Jack Lee*

2023 Summer reading under Dr. Shane D' Mello on a sheaf theoretic proof of de Rham isomorphism following *Griffiths and Harris* and *Bott and Tu*

2023 Winter reading under Dr. Vaibhav Vaish on representation theory of finite groups following *Fulton and Harris*

2024 Summer reading under Dr. Vaibhav Vaish on representation theory of Lie groups and Lie algebras following *Fulton and Harris*

2024 Symplectic manifolds and Hamiltonian flows following *Ana Cannas da Silva*

Seminars and workshops that I have attended

Graduate Students Seminar, IISER Mohali 2023-2026
Seminar IISER Mohali

Young mathematicians conference . . . . May 2024
Workshop IISER Mohali

Representation Theory of Quivers . . . . Winter 2024
Workshop IIT Kanpur

Geometric Aspect of Algebraic Varieties . . . . Spring 2025
Conference and celebration of mathematical legacy of Dr. Kapil Hari Paranjape IISER Mohali

Young mathematicians conference . . . . May 2025
Workshop IISER Mohali

Rigidity of Discrete Groups . . . . June 30-July 4, 2025
Workshop organized by Dr. Krishnendu Gongopadhyay and Dr. Pranab Sardar IISER Mohali

Tricks of the Trade Seminar . . . . Fall 2025
Seminar organized by Dr. Chentan Balwe IISER Mohali

Singapore-Bangalore "What is ...?" Workshop on Mathematics . . . . Fall 2025
Workshop organized by NUS Singapore and IISc Bangalore IISc Bangalore

Analytic group theory: amenability . . . . Fall 2025
Seminar organized by graduate students of IISER Mohali IISER Mohali

Young mathematicians conference . . . . May 2026
Workshop IISER Mohali

Teaching and outreach

2026 Volunteered in official help session on the theorem of Picard for second year students taking the course on ordinary differential equations in IISER Mohali in Spring 2026

2025 Volunteered to give a talk on Fourier series in a Science festival 2025 in IISER Mohali

2023 Volunteered in official help sessions on linear algebra for first year students in IISER Mohali in Fall 2023

2023-2026 Wrote a route-map for introductory resources for mathematics and physics

2022 Taught the geometry of curves and surfaces to second year students in IISER Mohali in Fall 2022

Achivements

2026 Selected for ALGANT Masters program in mathematics

2026 Qualified GATE 2026 in mathematics

Skills

English TOEFL iBT (November 2025) scored 99/120 (R: 25, L: 28, S: 24, W: 22)

Python Experience with libraries such as SnapPy and Sage.
Experience in NumPy, Sympy and Matplotlib for data analysis, algebra and visualization.

LaTeX Proficient in creating complex documents, including academic papers and presentations.

Notetaking I use Obsidian for taking notes on mathematics. I have experience in organizing and structuring information effectively, spread over hundreds of notes, using this tool.

Vector art 10 years of experience in vector art design using Adobe Illustrator such as my MS thesis poster

Image editing Familiar with Adobe Photoshop

Programming Familiar with C, C++, MATLAB, R, HTML, CSS, Javascript, VS code, Linux